Gradient Decoupling Guided Network for High-Resolution Remote Sensing Segmentation

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***Abstract*— For the semantic segmentation of remote sensing images, most existing methods focus on directly fusing unrefined low-level features with high-level features to enhance feature representation. However, these methods often neglect the potential feature entanglement within low-level features, making it challenging to accurately extract and restore spatial details. In this article, a gradient decoupling guided network (GDGNet) is proposed to alleviate this issue. The key components of GDGNet include the hybrid gradient enhancement (HGE) module, the hierarchical gradient attention (HGA) module, and the global–local context fusion (GLCF) module. First, the HGE aggregates learnable gradient convolutions to encode gradient information, enhancing the gradient features of low-level features. Then, the HGA reweights gradient decoupling masks (GDMs) to disentangle low-level features, guiding the network to focus on essential gradient regions. Finally, the GLCF fuses low- and high-level features, generating local and global contextual features and concatenating them to achieve segmentation. We conducted comparison and ablation experiments on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam datasets. The experimental results demonstrate the superiority of the proposed GDGNet over several state-of-the-art methods. The codes will be available at https://github.com/wangkaiwh331/GDGNet**

***Index Terms*— Attention module, gradient decoupling, gradient information enhancement, remote sensing image, semantic segmentation.**

I. Introduction

With the rapid development of satellite and aerial remote sensing technologies [1], high-resolution remote sensing images can be efficiently obtained. Semantic segmentation of remote sensing images aims to assign category labels to each pixel, and it plays a crucial role in various application scenarios such as land cover mapping [2], [3], environmental monitoring [4], [5], and disaster assessment [6], [7]. High-resolution remote sensing images provide rich spatial details for finer semantic segmentation of target scenes, while their complex land cover types and multiscale phenomena also render semantic segmentation more challenging.

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In recent years, deep learning methods [8], [9], [10] have achieved significant success in remote sensing image segmentation due to their powerful nonlinear modeling and contextual awareness capabilities. The fully convolutional network (FCN) [11] utilizes convolutional neural networks (CNNs) to extract features, achieving end-to-end training and dense prediction, resulting in significant breakthroughs in semantic segmentation tasks. Subsequently, strategies such as multiscale feature aggregation [12], [13], attention mechanism [14], [15], [16], and context extraction [17], [18], [19] have been developed, significantly enhancing the ability to capture multiscale features and contextual information in high-resolution remote sensing imagery. Despite this success, these methods often suffer from losing critical details such as boundaries and textures during continuous downsampling, making it challenging to recover image details [20].

The gradient describes the variation of pixel grayscale values at a certain point in an image and can represent boundary and texture features [21]. Due to its insensitivity to changes in color, position, and lighting, it exhibits good robustness in different scenarios. Based on the advantages of gradients, researchers have introduced gradient enhancement strategies during the extraction of low-level features to enhance the details in remote sensing images. For instance, Xiao et al. [22] designed directional convolutions composed of three strips of pixels to strengthen gradient information at boundaries. Additionally, Zhao et al. [23] derived pixel gradients by solving the derivative of the Gaussian function to emphasize the edges of objects in images, thereby improving object localization. Sun et al. [24] designed a boundary attention module that inputs the Canny edge feature map into the attention layer to enhance boundary attention. Additionally, some studies have drawn on traditional gradient operators to design gradient convolution kernels for extracting gradient features, thereby enhancing the representation of boundary information [20], [25], [26], [27].

Although these methods effectively mitigate the loss of image details, these gradient enhancement strategies still suffer from two critical challenges. On the one hand, these methods primarily focus on enhancing the gradient of object boundaries to improve boundary segmentation accuracy while not paying sufficient attention to the texture information of the surface features. Since different ground objects possess unique surface textures, extracting texture information could enhance the contextual consistency among objects of the same class [28]. Therefore, texture information is also valuable for

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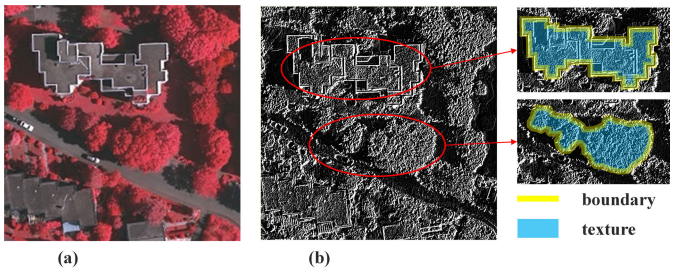


Fig. 1. Visualization of gradient maps extracted by the Sobel operator from the ISPRS Vaihingen dataset. (a) Original image. (b) Gradient image.

remote sensing image segmentation and needs more attention. On the other hand, existing methods often focus on directly fusing unrefined low-level features with high-level features to enhance feature representation, which may neglect the potential feature entanglement within low-level features. Our research finds that information such as image boundaries and surface textures may be entangled within the same low-level feature map. As shown in Fig. 1(a), the boundaries of objects such as buildings and trees in the original image are clear, and textures are visible within the objects. In the gradient image obtained by the Sobel operator [21] [as shown in Fig. 1(b)], pixels in the regions of boundaries and textures exhibit higher grayscale values, with boundaries typically appearing as continuous lines and textures as densely and uniformly distributed areas of higher brightness pixel points. Conversely, regions without boundaries or textures (e.g., roads and shadows) have lower grayscale values. Therefore, gradient images could highlight boundaries and textures based on the pixel grayscale values. However, in many cases, the pixel intensities of boundaries and textures in the gradient image are similar and difficult to distinguish. Thus, decoupling gradient information to differentiate between various boundary or texture features remains a challenge.

This article addresses the issue of insufficient texture information extraction by designing the hybrid gradient enhancement (HGE) module. This module integrates learnable Sobel and Laplacian of Gaussian (LoG) gradient convolutions. The Sobel convolution is employed to extract multidirectional boundary gradients, while the LoG convolution is used to extract surface texture gradients, thereby enhancing the representation of gradient information within the low-level features. To address the issue of gradient feature entanglement, we designed the gradient decoupling masks (GDMs) and the hierarchical gradient attention (HGA) module. The GDMs separately record the distribution of regions with different gradient features, such as the boundaries of buildings and cars, regularly patterned rooftops, rough tree canopies, and smooth road surfaces and shadows, preserving information from one region type while suppressing others. The HGA module adaptively adjusts the weights of the GDMs according to the input image content, guiding the network to focus on the information from the boundary or certain texture regions. Furthermore, a dual-branch global–local context fusion (GLCF) module is proposed to fuse low- and high-level features adaptively. This module uses multihead attention for global context extraction and uses HGE with standard convolution mixed convolutions to capture local details, allowing the model to leverage global context information without losing local detail features.

Based on the above considerations, we propose a gradient decoupling guided network (GDGNet). The main contributions can be summarized as follows.

1) An HGE module is proposed to enhance gradient information. The HGE module models traditional Sobel and LoG operators as learnable gradient convolutions, adaptively encoding gradient information to improve the boundary and texture features representation.
2) An HGA module is proposed for disentangling gradient information within low-level features. This module utilizes GDMs to adaptive decouple gradient features, aiding the model in focusing on the details of boundaries and texture regions.
3) A GLCF module is introduced, designed to adaptively fuse low- and high-level features and extract global and local contextual information, making features more discriminative.
4) A novel GDGNet is proposed for high-resolution remote sensing image segmentation. Experimental results on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam datasets demonstrate that our method achieves superior performance compared to state-of-the-art approaches.

The remainder of this article is organized as follows. Section II presents the related work, Section III introduces the GDGNet framework and related components, and Section IV presents the experimental settings, ablation studies, and comparison results. Finally, the conclusions of this article are drawn in Section V.

II. Related Works

A. Semantic Segmentation of High-Resolution Remote Sensing Images

In recent years, inspired by the successful advancements of deep learning in computer vision, CNNs have been extensively utilized for semantic segmentation tasks in remote sensing imagery. FCN [11] is a pioneering and successful attempt to use CNN for semantic segmentation, achieving end-to-end training and prediction for image segmentation at any resolution. Variants of FCN have shown good performance when applied to remote sensing tasks [29], [30], [31]. However, these methods often result in a loss of spatial resolution during continuous downsampling, making it challenging to recover fine image details. Furthermore, their limited receptive field is inadequate for handling the scale variability in remote sensing imagery.

In order to address these challenges, some research has focused on constructing networks based on multiscale feature aggregation [32], [33], [34]. Abdollahi et al. [35] employed a U-shaped architecture to aggregate multiscale features and utilized skip connections to mitigate the loss of details and spatial resolution caused by downsampling. Drawing from the residual learning framework, Yu et al. [36] applied the pyramid pooling module [37] to extract multiscale feature

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maps. Nogueira et al. [38] and Liu et al. [39] utilized dilated convolutions to increase receptive fields, enhancing contextual information for feature aggregation. Subsequently, Li et al. [40] designed a self-smoothing atrous convolution (SS-AConv) to mitigate grid artifacts caused by dilated convolution and integrate it into a cascaded multiscale structure to improve multiscale feature extraction. Zhang et al. [41] proposed a multiscale contextual aggregation network. It uses scale-aware information aggregation modules to expand branch receptive fields and conducts bottom-up feature fusion across the low-to-high branches, thereby integrating multiscale context information. Peng et al. [42] developed a cross fusion network (CF-Net), which captures more accurate small-scale semantic information by selecting informative features and enlarging the receptive field of low-level feature maps, improving the segmentation accuracy of small-scale objects. Li et al. [43] proposed a multiattention network (MANet), which leverages multiple efficient attention modules to capture contextual dependencies. Li et al. [44] developed an attention aggregation feature pyramid network (A^2-FPN), leveraging a feature pyramid network (FPN) to capture multiscale features and integrating an attention aggregation module to improve multiscale feature representation. In MCCANet, Zheng et al. [45] introduced a multiscale channelwise cross attention, dynamically and adaptively integrating features with context scale perception, effectively focusing on objects of various sizes.

Moreover, some works are devoted to extracting global contextual information to model the interrelatedness of pixels in different regions, thereby enhancing multiscale feature representation. To overcome the limitation of limited perception of long-range contexts caused by image cropping operations on large-scale RSIs, Ding et al. [46] proposed a wide-context network (WiCoNet) featuring two CNN branches for extracting features from local and global image levels. A context transformer is introduced to enhance local features' awareness of broad contextual information. Zhou et al. [47] proposed a hierarchical context network (HCN) that learns contextual information from pixel-to-pixel (P2P) and pixel-to-object (P2O) relationships. It extracts spatial details through P2P and semantic features through P2O. Li et al. [48] explore extracting and aggregating cross-layer contextual information by quantifying and representing the correlation between pixels and superpixels, developing an HCANet. Zhang et al. [49] designed an efficient shallow–deep interaction network (ESDINet), which can quickly extract spatial details and the global context of images through the detailed branch and semantic branch. Zuo et al. [50] combined the sparse spatial sampling strategy with the long-range relationship modeling capability to design the multiscale deformable attention net (MDANet), utilizing the deformable attention module (DAM) to adaptively capture neighborhood structural information for each pixel. Zhang et al. [51] introduced a truncation compensation network (TCNet), which predicts spatial positions of relevant categories, establishes spatial mappings for each category to enhance global semantic features, and integrates global features into local ones through a GLCF module context fusion module (CFM), effectively utilizing both global context and local fine-grained structures.

Furthermore, the transformer has shown great potential in modeling distance-free contextual information and has been successfully applied to RSI segmentation [52], [53]. Based on the global modeling capability of Swin Transformer [54], He et al. [55] developed the ST-UNet network, which utilizes the relational aggregation module to hierarchically integrate the global dependencies of Swin Transformer into the features of CNN. Wang et al. [56] introduced a densely connected Swin (DC-Swin), which employs the Swin Transformer as the backbone to extract the context information and utilizes a decoder with a densely connected feature aggregation module to restore the resolution. In order to capture rich global and local contextual details, Li et al. [57] proposed a typical encoder that hierarchically interlaces convolution and transformer, extracting local patterns and long-range dependencies through interleaved Swin Transformer and convolution blocks. Similarly, Liu et al. [58] proposed a global–local transformer segmentation framework (GLOTS), which utilizes transformer architecture to encode and decode input data, with a global–local attention module to extract global and local context information. Wang et al. [59] proposed the bilateral attention network (BANet), which employs a transformer backbone with multihead self-attention [60] to capture long-distance dependencies and convolutional operations to extract texture features. Similarly, based on the transformer, Wang et al. [61] introduced a global–local attention mechanism to model global and local information in the decoding stage of the proposed UNetFormer, leveraging global modeling capabilities while retaining local details.

Although the above methods have improved remote sensing image segmentation, they focus on aggregating diverse multiscale features and richer contextual information, often neglecting the potential feature entanglement within low-level features. Existing methods tend to fuse unrefined low-level features with high-level features to enhance detail representation, which may not be the optimal strategy. In contrast, GDGNet focuses on decoupling gradient information to generate GDMs, which guide the network's attention toward critical boundary and texture regions, thereby refining the low-level features to produce more precise spatial details.

B. Semantic Segmentation With Gradient Features

The boundaries and textures with larger gradient values are crucial information in image semantic segmentation tasks, playing an important role in reconstructing the local details of the object. Early researchers designed gradient detection operators based on first-order or second-order derivatives (e.g., Sobel and Feldman [21], Prewitt [62], LoG [63], and Canny [64]) to extract image gradients and successfully applied them to image boundary detection. Subsequently, these gradient detection operators were combined with deep learning methods and applied to image segmentation. For example, Hu et al. [25] first combined the Sobel algorithm with an FCN, designing an end-to-end learnable Sobel heuristic kernel for extracting boundary features. Similarly, Hou et al. [20] proposed a boundary-sensitive network (BSNet), where dynamic mixed gradient convolutional kernels based on the

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Sobel operator were designed to explicitly encode boundary information, enhancing semantic segmentation boundaries. Inspired by the Roberts operator, Zhou et al. [26] designed the difference-pooling module, which extracts edge gradient features by calculating the difference between the max-pooling layer and the average-pooling layer, enabling the extraction of edges in multiple directions. Wu et al. [65] designed a Canny enhanced high-resolution neural network (C-HRNet), where Canny edge features are utilized to enhance the prediction capabilities of boundaries. Shan and Jia [27] introduced a convolution-based probability gradient loss, which utilizes Sobel-like convolution kernels to extract ground truth and predicted probability gradients, enhancing accuracy near the object's boundary by maximizing the similarity between these two gradients. In addition, some researchers have introduced edge convolution to enhance image edge gradient features. For instance, to enhance gradient changes perpendicular to boundaries, Xiao et al. [22] designed a directional convolution composed of three-pixel stripes, where pixels within each stripe share identical weights. This approach smooths gradients parallel to the kernel direction while suppressing interference from features parallel to the boundary. Similarly, Zhao et al. [23] proposed a deep edge convolution-based method that extracts image pixel gradients by computing the derivative of the Gaussian function, thereby enhancing the edge information of image objects and improving the model's localization capability.

However, the abovementioned research mainly focuses on the boundary information in gradients but does not give adequate attention to the utilization of the texture gradients. Additionally, researchers utilize gradient histograms to extract texture features for image segmentation [66], [67]. For instance, Zhang et al. [28] applied texture features to remote sensing image segmentation, utilizing an improved 2-D gradient histogram and the minimum mean absolute deviation model for optimal threshold segmentation. This algorithm achieves good segmentation results for remote sensing images with Gaussian and Laplacian distribution histogram features. However, gradient histograms only contain intensity and directional information of the image, lacking pixel spatial positions, which hinders the representation of spatial distribution characteristics of the objects.

In addition, while using the gradient operator or gradient convolution to enhance boundary gradients, the studies above also extract and highlight some texture gradients, leading to the potential entanglement of boundary and texture information within low-level features, which may affect the extraction and restoration of spatial details. Some studies have utilized wavelet transform [68], [69] and Fourier transform [70] to decompose high-frequency (e.g., boundaries and complex textured regions) and low-frequency (e.g., smooth textured regions) components of images to help the model capture both detail information and overall structure, proving that separating boundary and texture information in images is an effective way to accurately identify spatial details. In this article, we leverage the differences between boundary and texture gradients to design the GDMs, which decouple the boundary and texture components at various levels from the low-level features, guiding the model to learn features from specific boundary and texture regions.

In summary, gradient features could effectively enhance the semantic segmentation of remote sensing images. However, few studies have focused on the differences in gradient features. Thus, utilizing various boundary and texture gradient information to refine model predictions for more accurate details remains challenging. Unlike existing gradient enhancement methods, GDGNet simultaneously models gradients of both ground objects' boundaries and surface textures, thereby enhancing boundary and texture information representation. Moreover, GDGNet introduces the HGA modules to guide the model in adaptively identifying important boundary and texture regions for extracting deep features, thereby maximizing the utility of gradient features in image segmentation.

III. Method

In this section, the architecture and the component modules of GDGNet will be introduced in detail. First, the overall architecture of GDGNet is presented in Section III-A. Then, the details of the HGE, GDM, HGA, and GLCF modules are introduced in Sections III-B–III-E. Finally, the loss functions used for network training are outlined in Section III-F.

A. Network Architecture

The overall architecture of GDGNet is illustrated in Fig. 2. It consists of five stages: encoding, gradient enhancement, GDMs generation, gradient attention, and decoding. The encoding stage employs ResNet18 as the backbone network to extract hierarchical features. This backbone comprises a 7×7 convolutional layer, a max-pooling layer, and four residual block (ResBlock) stages. The four ResBlock stages output feature maps F^{res} of varying resolutions: $H/4 \times W/4$, $H/8 \times W/8$, $H/16 \times W/16$, and $H/32 \times W/32$ (where H represents the height and W represents the width). The output feature maps F^{res} are generated as

$$F^{\text{res}} = f^{\text{en}}(I) \tag{1}$$

where I represents the input image and $f^{\text{en}}(\cdot)$ denotes the operation of the encoder. The gradient enhancement stage aims to enhance the boundary and texture gradient information within F^{res}, alleviating the insufficient extraction of texture information. This stage utilizes four HGE modules to output the feature F^{hge}. F^{hge} is calculated as

$$F^{\text{hge}} = \text{HGE}\left(F^{\text{res}}\right) \tag{2}$$

where $\text{HGE}(\cdot)$ represents the HGE operation.

Meanwhile, in the GDMs generation stage, the input image undergoes gradient decoupling to obtain four sets of GDMs, which record the spatial distribution of various boundary and texture regions within the hierarchical features.

Subsequently, in the gradient attention stage, HGA assigns weights to the gradient decoupling submasks (GDSMs) of each GDM and multiplies these GDSMs with the corresponding F^{hge} to emphasize the features of the critical regions, such as the boundaries of buildings and cars, the tree canopies, the

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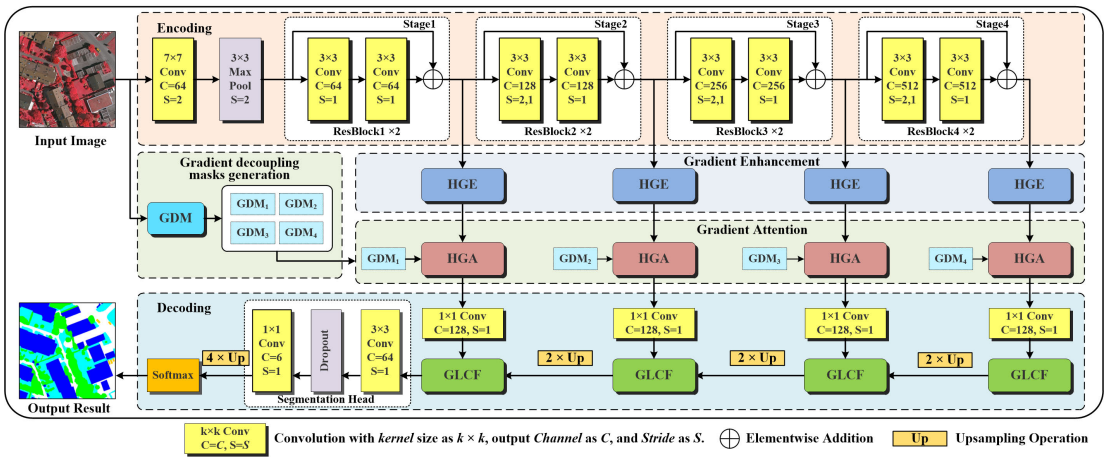


Fig. 2. Overall architecture of the GDGNet.

road surfaces, and the shadows, resulting in the HGA feature F^{hga}

$$F^{\text{hga}} = \text{HGA}\left(F^{\text{hge}}\right) \tag{3}$$

where HGA$(\cdot)$ represents the HGA operation. At this stage, the entanglement within the low-level features would be alleviated.

For the decoder, the four HGAs' output features are first unified to 128 channel dimensions using the 1×1 convolution. Then, these features are fused with the corresponding feature maps of the decoder by the GLCF module, extracting global and local contextual features. Finally, the output of the last GLCF module is fed into the segmentation head for prediction. The decoding process can be formulated as follows:

$$\begin{cases} F_i^{\text{glcf}} = \text{GLCF}\left(\text{Conv}_{1\times1}\left(F_4^{\text{hga}}\right)\right), & (i=1) \\ F_i^{\text{glcf}} = \text{GLCF}\left(\text{Conv}_{1\times1}\left(F_{5-i}^{\text{hga}}\right), \text{Up}_{2\times}\left(F_{i-1}^{\text{glcf}}\right)\right), \\ \qquad\qquad (i=2,3,4) \end{cases} \tag{4}$$

$$I^{\text{out}} = \text{Softmax}\left(\text{Up}_{4\times}\left(\text{Conv}_{1\times1}\left(\text{Drop}\left(\text{Conv}_{3\times3}\left(F_4^{\text{glcf}}\right)\right)\right)\right)\right) \tag{5}$$

where GLCF$(\cdot)$ represents the GLCF operation, F^{glcf} is the GLCF output, Drop$(\cdot)$ denotes the dropout operation, $\text{Conv}_{1\times1}(\cdot)$ denotes a 1×1 convolution operation, $\text{Conv}_{3\times3}(\cdot)$ denotes a 3×3 convolution operation, $\text{Up}_{4\times}(\cdot)$ denotes the $4\times$ upsampling operation, and I^{out} represents the final output.

B. HGE Module

Boundaries and textures are both crucial information in remote sensing images. The objective of this study is to enhance both by encoding gradients in low-level features. Therefore, we designed the HGE module to extract the boundary and texture information with sharp gradient changes.

Inspired by the wisdom of traditional gradient detection operators, we introduced two learnable gradient convolution kernels into the HGE module: Sobel and LoG convolution kernel. The Sobel operator excels at extracting edge information in four directions: 0°, 45°, 90°, and 135°. It performs neighborhood averaging on the image using a 3×3 kernel and then calculates edge points using the first-order derivative. Unlike the Sobel operator, the LoG operator does not depend on the gradient direction and can better highlight texture details in the image. It first applies Gaussian smoothing and then detects image gradient information based on the zero crossing of the Laplacian second-order derivative. Therefore, the HGE module employs four Sobel convolutions to extract boundary gradients from four directions and one LoG convolution to capture texture gradient information. According to the gradient rule of Sobel and LoG kernels, the learnable gradient kernels are designed, as shown in Fig. 3. The learnable 3×3 Sobel kernel corresponding to 0°–135° and the learnable 5×5 LoG kernel can be described as

$$K_{\text{Sob0}^\circ} = \begin{bmatrix} -a & -2a & -a \\ 0 & 0 & 0 \\ a & 2a & a \end{bmatrix} \tag{6}$$

$$K_{\text{Sob45}^\circ} = \begin{bmatrix} -2b & -b & 0 \\ -b & 0 & b \\ 0 & b & 2b \end{bmatrix} \tag{7}$$

$$K_{\text{Sob90}^\circ} = \begin{bmatrix} -c & 0 & c \\ -2c & 0 & 2c \\ -c & 0 & c \end{bmatrix} \tag{8}$$

$$K_{\text{Sob135}^\circ} = \begin{bmatrix} 0 & d & 2d \\ -d & 0 & d \\ -2d & -d & 0 \end{bmatrix} \tag{9}$$

$$K_{\text{LoG}} = \begin{bmatrix} 0 & 0 & e & 0 & 0 \\ 0 & e & 2e & e & 0 \\ e & 2e & -16e & 2e & e \\ 0 & e & 2e & e & 0 \\ 0 & 0 & e & 0 & 0 \end{bmatrix} \tag{10}$$

where a, b, c, d, and e are the trainable parameters.

The structure of the HGE module is shown in Fig. 4. The input feature map F^{res} is first passed through the five gradient convolutional layers mentioned above in parallel to extract gradient features. Each gradient convolution has a stride of 1, and the output channels are the same as those of F^{res}. Subsequently, these five output features are passed

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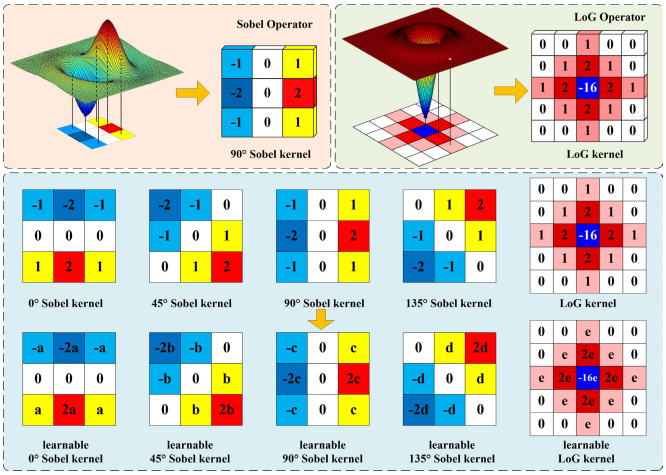


Fig. 3. Learnable gradient kernels. The learnable gradient kernels are designed based on each Sobel and LoG kernel.

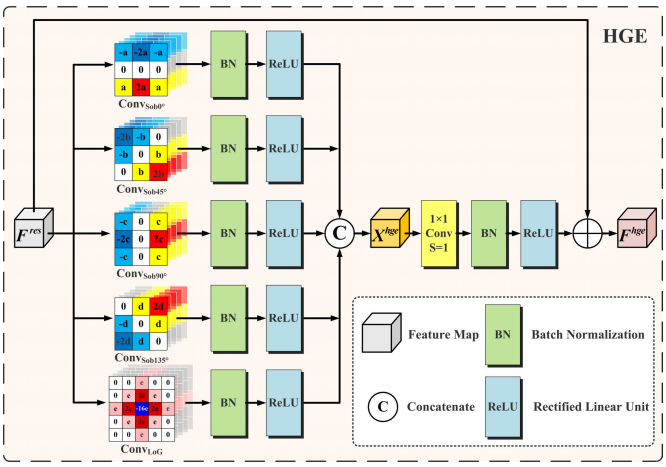


Fig. 4. Illustration of the HGE module.

through the batch normalization (BN) layer, rectified linear unit (ReLU) activation function, and concatenated along the channel dimension to obtain X^{hge}. Then, a 1×1 convolution is applied to adjust the channel size of X^{hge} to match that of F^{res}. Finally, the resulting feature map is added to F^{res} using skip connections to obtain the gradient-enhanced output feature F^{hge}. The computation process can be expressed as

$$X^{\text{hge}} = \text{Concat}\left(\left\{\text{ReLU}\left(\text{BN}\left(\text{Conv}_i^{\text{grad}}\left(F^{\text{res}}\right)\right)\right)\right\}_{i=1}^{5}\right) \quad (11)$$

$$F^{\text{hge}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}\left(X^{\text{hge}}\right)\right) + F^{\text{res}}\right) \quad (12)$$

where $\text{Conv}_i^{\text{grad}}(\cdot)$ represents the gradient convolutional operation (for i ranging from 1 to 5, corresponding to convolutional kernels $K_{\text{Sob0°}}$, $K_{\text{Sob45°}}$, $K_{\text{Sob90°}}$, $K_{\text{Sob135°}}$, and K_{LoG}), $\text{BN}(\cdot)$ is the batch normalization, $\text{ReLU}(\cdot)$ is the rectified linear unit, and $\text{Concat}(\cdot)$ denotes the concatenate operation.

C. GDM Generation

As previously discussed, the boundary and texture features within the low-level features may still be intertwined, which may confuse when distinguishing semantics. To address this issue, we designed GDMs to capture the spatial distribution of various boundary and texture gradients. GDMs are utilized as spatial attention maps and multiplied element-wise with low-level features to highlight specific region features while suppressing irrelevant ones, achieving the decoupling of low-level features. The construction of GDMs involves two steps: gradient decoupling and GDM construction, as described next.

1) *Gradient Decoupling:* To decouple gradient information, we adopted a threshold segmentation method based on the dynamic threshold to separate the gradient map into multiple binary images, each indicating different texture regions. Fig. 5 illustrates an example from the Vaihingen dataset. First, the LoG convolution was utilized to obtain the gradient map and construct the gradient histogram [as shown in Fig. 5(c)]. Subsequently, starting from zero gradient, the cumulative pixel count along the horizontal axis of the histogram is calculated, and the ratio of the cumulative pixel count to the total pixel count is defined as p ($p \in [0\%, 100\%]$). Subsequently, thresholds $[p_i, p_{i+1}]$ are set for threshold segmentation to capture pixels with gradient values between p_i and p_{i+1} (where $i = 1, 2, \ldots, n$, and $p_i < p_{i+1}$; in this article, the GDM consists of five GDSMs, and hence, n is set to 5). Fig. 5(d) shows the binary images of segmentation results for each threshold interval using (0%, 20%, 40%, 60%, 80%, and 100%) as an example. For example, the segmentation results corresponding to the [0%, 20%] interval depict shadow regions, the [60%, 80%] interval mainly represents the distribution of trees, and the [80%, 100%] interval primarily highlights the boundaries of buildings or the distribution of cars. These results demonstrate that the spatial distribution of pixels extracted through this threshold segmentation method corresponds to texture regions of different roughness, effectively decoupling the image gradients.
2) *GDM Construction:* First, local averaging operations are applied to the segmented binary images to convert the sparsely distributed pixels into density maps representing the continuous spatial distribution of specific gradient intensities. Then, the pixel values in the density maps are normalized to [0, 1], where the values represent spatial attention weights. Finally, the normalized results are used as GDMs, yielding n GDSMs, as shown in Fig. 5(e).

Fig. 6 demonstrates the construction process of the GDMs. First, the original image $I \in \mathbb{R}^{H\times W\times 3}$ is downscaled to $H/2 \times W/2 \times 3$ by an average-pooling layer. Subsequently, the mean is computed along the channels to generate a grayscale age of size $H/2 \times W/2 \times 1$. Then, the gradient image $\text{GI} \in \mathbb{R}^{H/2\times W/2\times 1}$ is extracted using LoG convolution. The process can be formulated as follows:

$$\text{GI} = \text{Conv}_{\text{LoG}}(\text{Mean}(\text{AP}_{3\times3}(I))) \quad (13)$$

where $\text{AP}_{3\times3}(\cdot)$ denotes the 3×3 average-pooling operation, $\text{Mean}(\cdot)$ represents the operation of averaging along the channel dimension, and $\text{Conv}_{\text{LoG}}(\cdot)$ is the LoG convolution.

Subsequently, the gradient image GI is thresholded using five sets of thresholds $[p_i, p_{i+1}]$ ($i = 1, 2, 3, 4, 5$, where $p_1 = 0$ and $p_6 = 1$), resulting in five binary images.

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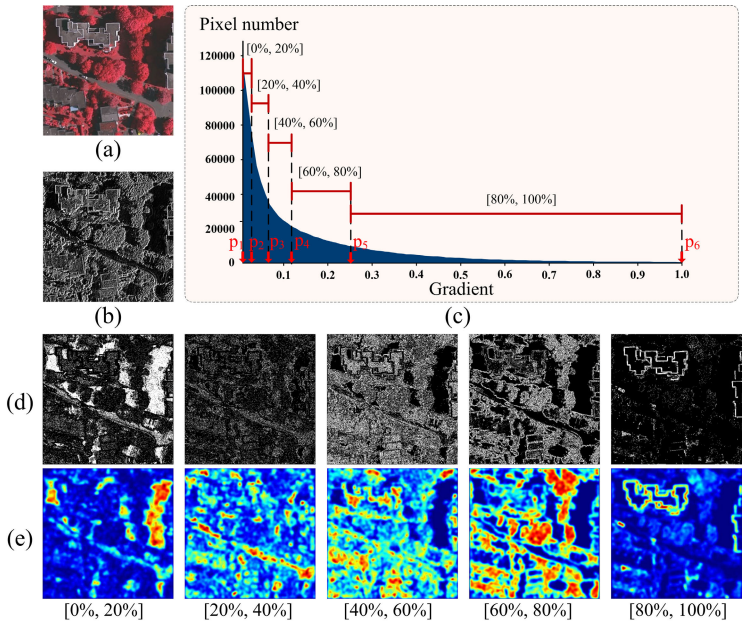


Fig. 5. Illustration of the GDM. (a) Original image. (b) Gradient image. (c) Gradient histogram. (d) Binary image resulting from threshold segmentation of gradient image. (e) GDSMs.

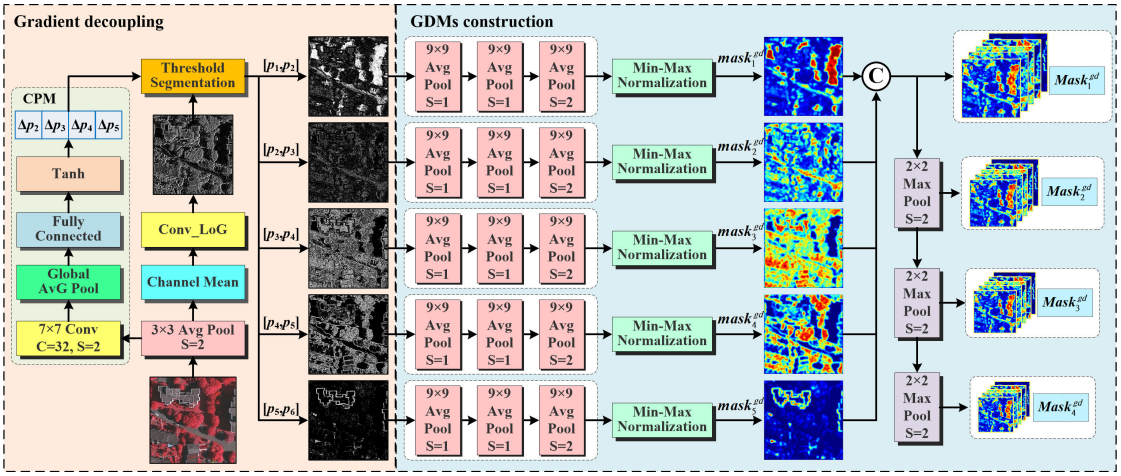


Fig. 6. Illustration of GDM construction.

Since the gradient information in different input images is variable, we propose a coefficient prediction module (CPM) to adaptively adjust the threshold parameters p_i, thereby enhancing the network's ability to decouple gradient features. The CPM generates adaptive coefficients for parameters p_2–p_5 according to the image content, as illustrated in Fig. 6. Initially, a convolutional layer is employed to learn the features of the input image. Subsequently, global average pooling is used to aggregate the global spatial information of these features, resulting in a feature vector. This feature vector is then mapped to a $1 \times 1 \times C$ (where $C = 4$, corresponding to the four adaptive coefficients) vector using a fully connected (FC) layer. The hyperbolic tangent function (Tanh) processes the $1 \times 1 \times C$ vector to obtain the weight coefficients Δp_i. The Tanh maps the adaptive coefficients to the range of $[-1, 1]$, which are then scaled by a coefficient η (in this study, η is set to 0.1) to adjust the magnitude of Δp_i. Finally, p_i is updated by adding Δp_i. These processes can be formulated as follows:

$$\Delta p = \text{Tanh}(\text{Linear}(\text{AP}_{H\times W}(\text{Conv}_{7\times 7}(\text{AP}_{3\times 3}(I))))) \quad (14)$$

$$p_i = p_i + \eta \Delta p_i, \quad (i = 2, 3, 4, 5) \quad (15)$$

where $\text{AP}_{H\times W}(\cdot)$ denotes the global average-pooling operation, $\text{Linear}(\cdot)$ represents an FC operation, and $\text{Tanh}(\cdot)$ is the hyperbolic tangent function. In this study, the initial p_2–p_5 values are set to 0.2, 0.4, 0.6, and 0.8, respectively.

After obtaining five binary images, each image is transformed into a density map through three 9×9 average-pooling layers. Finally, min–max normalization is applied to convert the pixel values of the density map into weights, generating GDSM $\text{mask}_1^{\text{gd}} \in \mathbb{R}^{H/4\times W/4\times 1}$, which are then concatenated to form the GDM $\text{Mask}_1^{\text{gd}} \in \mathbb{R}^{H/4\times W/4\times 5}$. $\text{Mask}_1^{\text{gd}}$ can be

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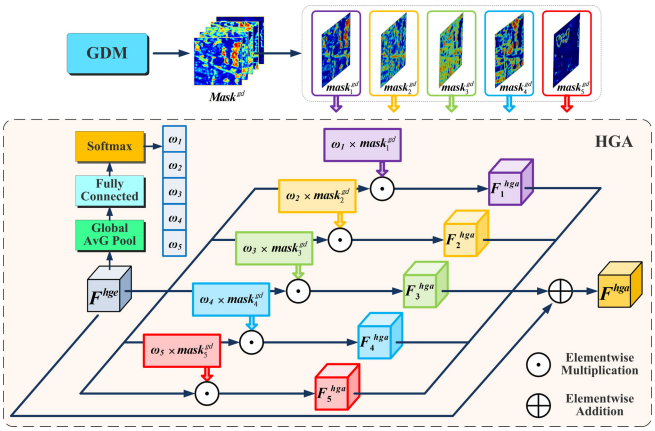


Fig. 7. Overall framework of HGA.

formulated as follows:

$$\mathrm{Mask}_1^{\mathrm{gd}} = \mathrm{Concat}\Big(\big\{\mathrm{Norm}\big(\mathrm{AP}_{9\times9}\big(\mathrm{AP}_{9\times9}\big(\mathrm{AP}_{9\times9} \times\big(\mathrm{TS}_{(p_i,p_{i+1})}(\mathrm{GI})\big)\big)\big)\big\}_{i=1}^{n}\Big) \quad (16)$$

where $\mathrm{TS}_{(p_i,p_{i+1})}(\cdot)$ denotes the threshold segmentation operation, obtaining pixels belonging to $[p_i, p_{i+1}]$. $\mathrm{Norm}(\cdot)$ represents the min–max normalization operation. Additionally, $\mathrm{Mask}_1^{\mathrm{gd}}$ is downsampled using max pooling to obtain $\mathrm{Mask}_2^{\mathrm{gd}} \in \mathbb{R}^{H/8\times W/8\times 1}$, $\mathrm{Mask}_3^{\mathrm{gd}} \in \mathbb{R}^{H/16\times W/16\times 1}$, and $\mathrm{Mask}_4^{\mathrm{gd}} \in \mathbb{R}^{H/32\times W/32\times 1}$.

D. HGA Module

Considering that the importance of different boundaries and textures varies, neglecting these differences could suppress critical features or overemphasize less significant ones, potentially affecting the model's performance. To address this, we designed the HGA module, which adaptively adjusts the weight coefficients of each GDSM in the GDM based on the content of the input image, enabling the network to extract more features from the regions corresponding to the GDSM with higher weights.

The framework of the HGA is illustrated in Fig. 7. First, $\mathrm{Mask}_i^{\mathrm{gd}}$ with the same size as the input feature map F^{hge} is generated. Then, global average pooling is utilized to convert F^{hge} into a feature vector v, which is mapped into a vector $\hat{v}$ with length n using an FC layer. Next, the feature vector $\hat{v}$ undergoes a softmax operation to weight coefficients ω_i, where the sum of ω_i is 1, deciding the contribution of $\mathrm{mask}_i^{\mathrm{gd}}$. Following this, the weights ω_i are multiplied by the $\mathrm{mask}_i^{\mathrm{gd}}$ and then multiplied by F^{hge}. The resulting product is added to F^{hge} to obtain the output feature F^{hga}. F^{hga} can be formulated as

$$\omega = \mathrm{Softmax}\big(\mathrm{Linear}\big(\mathrm{AP}_{H\times W}\big(F^{\mathrm{hge}}\big)\big)\big) \quad (17)$$

$$F^{\mathrm{hga}} = \sum_{i=1}^{5} \omega_i \mathrm{mask}_i^{\mathrm{gd}} \odot F^{\mathrm{hge}} + F^{\mathrm{hge}} \quad (18)$$

where $\odot$ represents the elementwise multiplication.

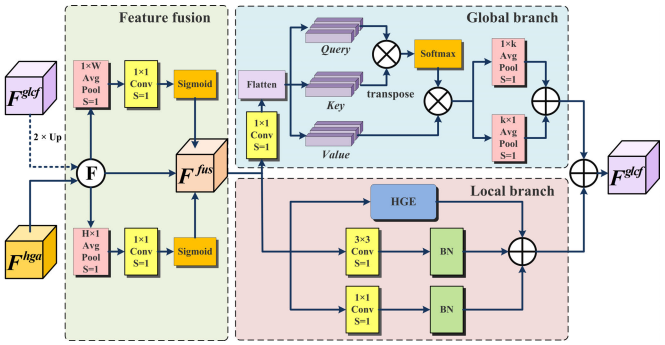


Fig. 8. Details of the GLCF.

E. GLCF Module

The low-level features preserve rich spatial details of urban scenes, while the deep high-level features provide precise semantic information. Although directly summing these two types of features is feasible, it may reduce segmentation accuracy [61], [71]. Inspired by the coordinate attention (CA) mechanism [72] and the Swin Transformer [54], we designed the GLCF module to fuse low- and high-level features, shrink the semantic gap between them, and effectively utilize local details and global information of the fusion features for further accuracy improvement. As illustrated in Fig. 8, the GLCF module consists of two stages: feature fusion and global–local context extraction. In the feature fusion stage, low- and high-level features are weighted and fused using a learnable parameter α. The weighted summation adaptively adjusts the weights according to the contribution of these two features, which can be expressed as

$$\begin{cases} X^{\mathrm{fus}} = F_i^{\mathrm{hga}}, & (i=4) \\ X^{\mathrm{fus}} = \alpha F_i^{\mathrm{hga}} + (1-\alpha)\mathrm{Up}_{2\times}\left(F_{4-i}^{\mathrm{glcf}}\right), & (i=1,2,3). \end{cases} \quad (19)$$

Then, long-range pooling is performed on F^{fus} along the horizontal and vertical directions using pooling kernels of size $1 \times W$ and $H \times 1$, respectively. This operation aggregates features along the two spatial directions, resulting in two direction-aware feature maps. These feature maps capture the long-range dependencies in one spatial direction while preserving the positional information in the other. Subsequently, the direction-aware feature maps are processed using 3×1 and 1×3 convolutions, respectively. The resulting features are then transformed into horizontal attention A^h and vertical attention A^w using the sigmoid function. Finally, these attention maps are element-wise multiplied by X^{fus} to generate F^{fus}. F^{fus} can be formulated as

$$A^h = \mathrm{Sigmoid}\big(\mathrm{Conv}_{1\times1}\big(\mathrm{AP}_{1\times W}\big(X^{\mathrm{fus}}\big)\big)\big) \quad (20)$$

$$A^w = \mathrm{Sigmoid}\big(\mathrm{Conv}_{1\times1}\big(\mathrm{AP}_{H\times1}\big(X^{\mathrm{fus}}\big)\big)\big) \quad (21)$$

$$F^{\mathrm{fus}} = A^h \times A^w \times X^{\mathrm{fus}} \quad (22)$$

where $\mathrm{Sigmoid}(\cdot)$ represents the sigmoid activation function.

Although the global context is crucial for the semantic segmentation of remote sensing images, local information remains essential for preserving rich spatial details. Therefore, in the

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global–local context extraction stage, two parallel branches are constructed to extract global and local contexts, as illustrated in Fig. 8.

In the global branch, we adopt a window-based multihead self-attention mechanism to extract long-range contextual information. First, a 1×1 convolution is employed to triple the number of channels in the input feature map. Subsequently, the window partition and flatten operation are applied to partition the feature map and flatten it into query ($Q \in \mathbb{R}^{((H/w)\times(W/w)\times h)\times(w\times w)\times(C/h)}$), key ($K \in \mathbb{R}^{((H/w)\times(W/w)\times h)\times(w\times w)\times(C/h)}$), and value ($V \in \mathbb{R}^{((H/w)\times(W/w)\times h)\times(w\times w)\times(C/h)}$) vectors. The window size w and the number of heads h are both set to 8. Following that, the Q, K, and V vectors are input into the multihead self-attention module to calculate the self-attention feature, formulated as follows:

$$(Q, K, V) = \text{Flatten}\left(\text{Conv}_{1\times1}\left(F^{\text{fus}}\right)\right) \tag{23}$$

$$F^{\text{sa}} = \text{Softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \times V \tag{24}$$

where d_k is the channel dimension of K.

Subsequently, the cross-shaped window context interaction method [61] is employed for nonoverlapping local window information exchange, formulated as follows:

$$F^{\text{global}} = \text{AP}_{1\times k}\left(F^{\text{sa}}\right) + \text{AP}_{k\times1}\left(F^{\text{sa}}\right). \tag{25}$$

In the local branch, local features are extracted in parallel using one 1×1 convolution layer, one 3×3 convolution layer, and an HGE module. Two BN operations are subsequently applied before the addition operation. The output local feature F^{local} can be formulated as

$$F^{\text{local}} = \text{BN}\left(\text{Conv}_{1\times1}\left(F^{\text{fus}}\right)\right) + \text{BN}\left(\text{Conv}_{3\times3}\left(F^{\text{fus}}\right)\right) + \text{HGE}\left(F^{\text{fus}}\right). \tag{26}$$

Finally, F^{local} is added to F^{global} to obtain F^{glcf}

$$F^{\text{glcf}} = F^{\text{global}} + F^{\text{local}}. \tag{27}$$

F. Loss Function

In the training phase, we used the Dice loss $\mathcal{L}_{\text{ce}}$ and cross-entropy loss $\mathcal{L}_{\text{dice}}$ to train the entire network. The loss $\mathcal{L}$ can be formulated as

$$\mathcal{L}_{\text{ce}} = -\frac{1}{M}\sum_{m=1}^{M}\sum_{p=1}^{P} y_p^{(m)} \log \hat{y}_p^{(m)} \tag{28}$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2}{M}\sum_{m=1}^{M}\sum_{p=1}^{P} \frac{\hat{y}_p^{(m)} y_p^{(m)}}{\hat{y}_p^{(m)} + y_p^{(m)}} \tag{29}$$

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{dice}} \tag{30}$$

where M and P denote the number of samples and the number of categories, respectively; $y^{(m)}$ and $\hat{y}^{(m)}$ represent the one-hot encoding of the actual semantic labels and the corresponding softmax output of the network, respectively; and $\hat{y}_p^{(m)}$ represents the confidence of sample m belonging to category p.

IV. Experimental Results

A. Dataset

1) Vaihingen Dataset: The Vaihingen dataset consists of 33 orthophoto images with a spatial resolution of 9 cm and an average size of 2494 × 2064 pixels. Each image is composed of three channels: red, green, near infrared, and the corresponding digital surface model (DSM). All the images are labeled with six landcover categories: impervious surfaces, buildings, low vegetation, trees, cars, and background. In our experiments, we utilized images with the following IDs for training: 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 32, 33, and 37. Image 30 is used for validation, and the remaining images are used for testing. Note that the DSM is not used in our experiments.

2) Potsdam Dataset: The Potsdam dataset consists of 38 orthophoto images with a size of 6000 × 6000 pixels and a spatial resolution of 5 cm. Each image contains near-infrared, red, green, and blue bands, the corresponding DSM, and the normalized digital surface model (NDSM). As with the Vaihingen dataset, it includes impervious surfaces, buildings, low vegetation, trees, cars, and background, totaling six landcover categories. We used 14 images for testing, with IDs 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13. Image 2_10 is used for validation, and the remaining images are used for training. Similarly, only three bands (red, green, and blue) were employed in the experiments.

B. Experiment Settings

1) Experimental Environment: All experiments were conducted on a computer with a 2.40-GHz Intel Xeon E5-2680 v4 central processing unit (CPU), 64-GB RAM, and an NVIDIA TITAN Xp graphics processing unit (GPU). The PyTorch 1.9 framework served as the running framework, while Python 3.6 was employed as the programming environment.

2) Data Preprocessing: The original images are cropped into patches with the size of 512 × 512 pixels. During the training phase, the patch sizes are randomly adjusted using five scaling factors: 0.5, 0.75, 1.0, 1.25, and 1.5. Subsequently, random vertical flip, random horizontal flip, and random rotation augmentations are applied to the image patches. During the testing phase, random flip augmentations are used.

3) Implementation Details: During the training process, the AdamW optimization algorithm is utilized for all models in the experiments. The learning rate is set to 0.0006, and the weight decay value is 0.01. For the Vaihingen and Potsdam datasets, the number of training epochs is set to 200, and the batch size is 8. In all experiments, we use the Resnet-18 models as backbone networks and pretrained on the ImageNet dataset [73].

4) Evaluation Metrics: Three evaluation metrics are used to evaluate the proposed model, including overall accuracy (OA), the mean intersection over union (mIoU), and mean $F1$-score (m$F1$). OA represents the probability of correctly classifying all pixels and can be expressed as

$$\text{OA} = \frac{\sum_{k=1}^{N} \text{TP}_k}{\sum_{k=1}^{N} \text{TP}_k + \text{FP}_k + \text{TN}_k + \text{FN}_k} \tag{31}$$

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TABLE I
ABLATION STUDY OF EACH COMPONENT OF THE GDGNET ON TWO DATASETS

Dataset	Method	HGE	HGA	GLCF	mIoU(%)	mF1(%)	OA(%)
Vaihingen	Baseline	-	-	-	80.88	89.16	92.24
		√	-	-	82.51	90.19	92.70
		√	√	-	83.14	90.58	92.88
		√	√	√	**83.64**	**90.88**	**93.16**
Potsdam	Baseline	-	-	-	82.82	90.49	88.98
		√	-	-	83.96	91.16	89.76
		√	√	-	84.73	91.59	90.19
		√	√	√	**85.60**	**92.10**	**90.78**

TABLE II
COMPARATIVE EXPERIMENT RESULTS OF HGE AND ITS COUNTERPARTS ON THE VAIHINGEN DATASET

Method	mIoU (%)	mF1 (%)	OA (%)
Baseline	80.88	89.16	92.24
Baseline+HGE_CNN	82.08	89.92	92.45
Baseline+HGE_Sob	82.36	90.10	92.62
Baseline+HGE_LoG	82.46	90.16	92.56
Baseline+HGE1	82.45	90.16	92.65
Baseline+HGE2	82.50	90.20	92.60
Baseline+HGE	**82.62**	**90.26**	**92.73**

where TP_k, FP_k, TN_k, and FN_k indicate the number of true positive, false positive, true negative, and false negative pixels, respectively, for the object indexed as class k.

The mIoU represents the ratio of intersection and union of the true and predicted values, which is expressed as follows:

$$\mathrm{mIoU} = \frac{1}{N}\sum_{k=1}^{N}\frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k}. \tag{32}$$

The $F1$-score can be calculated based on completeness (recall) and correctness (precision) with the following formula:

$$F1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{33}$$

$$\mathrm{Precision} = \frac{1}{N}\sum_{k=1}^{N}\frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k} \tag{34}$$

$$\mathrm{Recall} = \frac{1}{N}\sum_{k=1}^{N}\frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k}. \tag{35}$$

C. Ablation Study

The proposed GDGNet contains key modules HGE, HGA, and GLCF. In order to assess the contributions of each module to the results, we conduct ablation experiments by adding one module at a time. Since the GDGNet is similar in structure to U-Net, we use the U-Net with a ResNet18 backbone as the baseline model, and the relevant results are reported in Table I.

1) Effectiveness of HGE: Table I shows that the HGE module improves the mIoU by 1.63%, m$F1$ by 1.03%, and OA by 0.46% on the Vaihingen dataset. Similarly, on the Potsdam dataset, the HGE module improves the mIoU by 1.14%, m$F1$ by 0.67%, and OA by 0.78%.

To assess the performance enhancement of HGE, we compared it with its counterparts using the Vaihingen dataset, shown in Fig. 9. We replaced the LoG and Sobel convolutional kernels in the HGE module with vanilla convolutional kernels and evaluated the performance of these networks. Furthermore, we examined different fusion methods for the LoG and Sobel convolutions outputs, with results detailed in Table II. Employing LoG and Sobel convolution leads to varying degrees of improvement in classification accuracy. The most notable improvement is observed when both LoG and Sobel convolutions are utilized, coupled with a concatenation method for feature fusion. In this situation, mIoU, m$F1$, and OA achieve their peak values of 82.62%, 90.26%, and 92.73%, respectively.

2) Effectiveness of HGA: As evident from Table I, the model's classification performance is further enhanced with the application of the HGA module. Specifically, the mIoU increased by 0.63% and 0.77% for the Vaihingen and Potsdam datasets, the m$F1$-scores increased by 0.39% and 0.43%, and the OA scores increased by 0.18% and 0.43%.

Furthermore, to further validate the effectiveness of the HGA module, we incrementally added the HGA module to the four output stages of the backbone network, as shown in Table III. In this experiment, stages without the HGA module mean that the features from these stages are not decoupled, and their internal features may still be entangled. Experimental results indicate that the network's predictive performance gradually improves as the HGA module is added. When the HGA module is added to all four stages, the model achieves the highest mIoU, m$F1$, and OA of 83.14%, 90.58%, and 92.88%, respectively. Compared to the baseline, the $F1$-score for the car category shows the largest improvement, increasing by 0.97%, while the $F1$-scores for the building, low vegetation, and tree categories also improved by 0.30%, 0.28%, and 0.32%, respectively. These results confirm that

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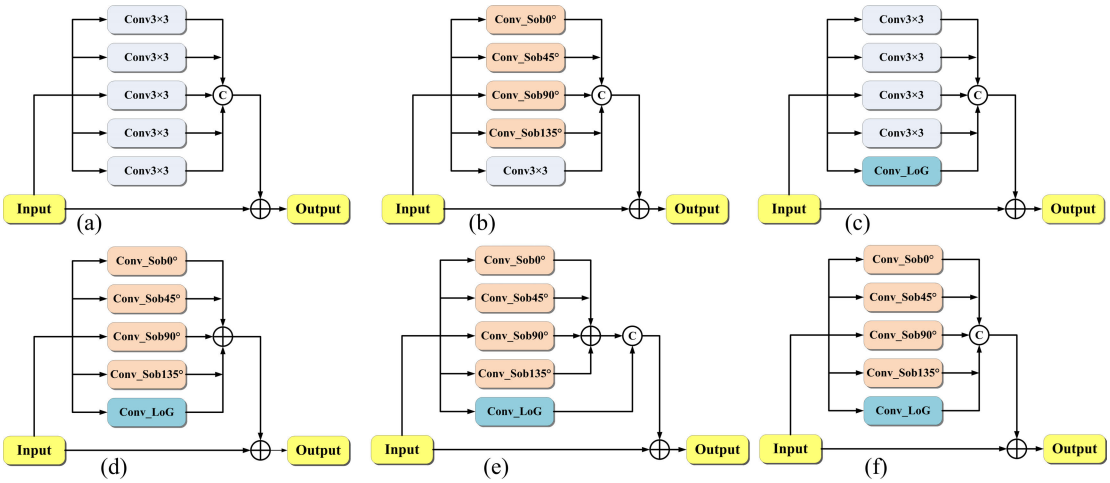


Fig. 9. Counterparts to compare against HGE. (a) HGE_CNN. (b) HGE_Sob. (c) HGE_Log. (d) HGE1. (e) HGE2. (f) HGE.

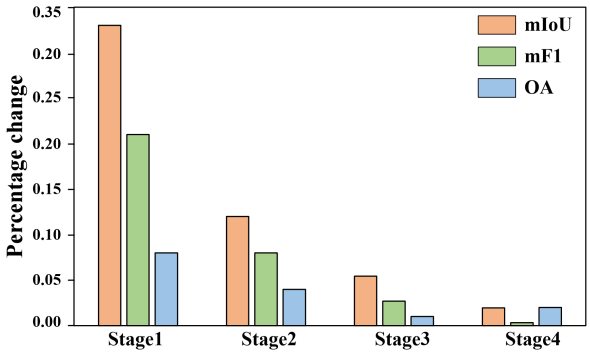


Fig. 10. Comparison of performance improvement on HGA at different stages.

the HGA module effectively alleviates feature entanglement in low-level features. As shown in Fig. 10, with the network deepening, the performance enhancement provided by HGA diminishes. This finding demonstrates that the network captures more detailed information at shallow layers.

Fig. 11 shows the prediction results and error maps of the baseline model and the baseline with the HGA added to the backbone's four output stages. As illustrated in the first row of Fig. 11, the prediction error on building surfaces is reduced after incorporating HGA, indicating that HGA could guide the network to focus on the texture features of object surfaces, ensuring consistent predictions for surface regions with similar textures. Similarly, in the second row of Fig. 11, the model with the HGA more accurately distinguishes between low vegetation and trees. Furthermore, as shown in the third row of Fig. 11, the model achieves more accurate predictions near object boundaries after adding the HGA, demonstrating the effectiveness of the HGA module in improving boundary prediction accuracy.

To validate the effectiveness of GDM in refining low-level features within the HGA module, we visualized the output features of a GDM. Taking ResBlock Stage 1 of the encoder as an example, Fig. 12 shows some visualization examples of the output features obtained without GDM and those generated by the five GDSMs contained in GDM. In Fig. 12(b), the highlighted pixel distribution in the original feature map generated by Stage 1 of the backbone network is relatively uniform, with no significant differences in pixel values between different classes. In contrast, GDM enhances features in specific regions while suppressing others. As shown in Fig. 12(c)–(g), in $GDSM_1$, shadow regions are highlighted, indicating that $GDSM_1$ could guide the model to focus on features within shadow regions. Additionally, $GDSM_2$ concentrates primarily on roads with smooth surfaces, while $GDSM_3$ emphasizes features related to building surfaces and grassland regions. $GDSM_4$ tends to capture rough-textured regions, such as trees, and the boundaries of buildings and cars are predominantly highlighted in $GDSM_5$. These results indicate that the HGA could flexibly leverage GDMs to refine low-level features, guiding the model to focus on various boundary and texture regions adaptively, thereby enhancing segmentation accuracy.

3) Effectiveness of GLCF: As shown in Table I, after introducing the GLCF module, the mIoU, m$F1$, and OA have increased by 0.50%, 0.30%, and 0.28% on the Vaihingen dataset, respectively. Similarly, there has been an increase of 0.87%, 0.51%, and 0.59% on the Potsdam dataset.

To investigate the performance of the global and local branches within the GLCF module, we conducted experiments with different settings, as detailed in Table IV. The results indicate that removing the global branch reduced the mIoU, m$F1$, and OA by 0.30%, 0.18%, and 0.11%, respectively. From Fig. 13(c), it is evident that due to the lack of global context, significant errors are observed within the building highlighted within the red box, demonstrating the importance of the global feature extraction branch. Contrarily, removing the local branch reduced the mIoU, m$F1$, and OA by 0.34%, 0.20%, and 0.20%, respectively. This decrease is attributed to the lack of local details, resulting in a noticeable decrease in the classification accuracy of small objects such as cars. As depicted by the blue box in Fig. 13(d), removing the local branch led to blurred boundaries for small objects.

D. Comparison With State-of-the-Art Methods

In this section, the proposed GDGNet is compared with some state-of-the-art semantic segmentation methods on the Vaihingen and Potsdam datasets. The comparison models

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TABLE III
ABLATION EXPERIMENT RESULTS ON HGA AT DIFFERENT STAGES ON THE VAIHINGEN DATASET

Method	Stage 1	Stage 2	Stage 3	Stage 4	F1 per category (%)					mIoU (%)	mF1 (%)	OA (%)
					ImpSurf	Building	LowVeg	Tree	Car			
Baseline + HGE	-	-	-	-	96.48	95.09	83.40	89.20	86.78	82.51	90.19	92.70
	√	-	-	-	96.52	95.25	83.53	89.39	87.66	82.95	90.47	92.81
Baseline + HGE + HGA	√	√	-	-	96.53	95.32	83.64	89.51	87.74	83.07	90.55	92.84
	√	√	√	-	**96.54**	95.38	83.68	89.51	**87.75**	83.12	90.57	92.86
	√	√	√	√	**96.54**	**95.39**	**83.69**	**89.52**	**87.75**	**83.14**	**90.58**	**92.88**

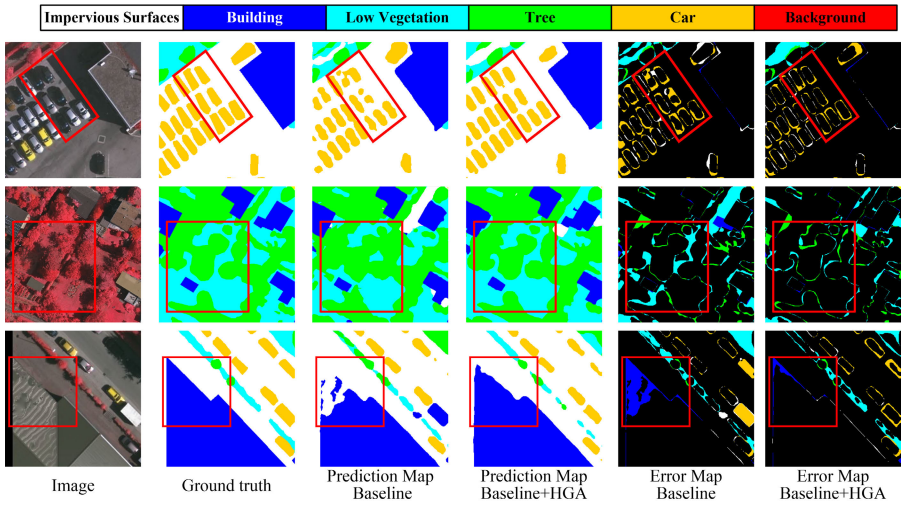


Fig. 11. Prediction results and error maps of the Baseline and Baseline + HGA. The black regions in the error maps represent the correct prediction.

TABLE IV
ABLATION EXPERIMENT RESULTS OF THE GLOBAL AND LOCAL BRANCHES IN THE HGA MODULE ON THE VAIHINGEN DATASET

Method	Global	Local	F1 per category (%)					mIoU(%)	mF1(%)	OA(%)
			ImpSurf	Building	LowVeg	Tree	Car			
GDGNet	√	-	96.72	95.46	84.02	89.56	87.76	83.34	90.70	93.05
	-	√	96.60	95.39	83.91	89.55	87.96	83.30	90.68	92.96
	√	√	**96.85**	**95.57**	**84.18**	**89.68**	**88.14**	**83.64**	**90.88**	**93.16**

include Deeplab V3+ [74], ABCNet [32], BSNet [20], MANet [43], A^2-FPN [44], MCCANet [45], DC-Swin [56], CMTFNet [52], BANet [59], and UnetFormer [61]. All the experiments are conducted under the same conditions.

1) Comparisons on the Vaihingen Dataset: Table V presents the segmentation performance of various comparative methods on the Vaihingen dataset. As shown in Table V, the traditional method DeepLab V3+ designed for natural image segmentation exhibits the lowest classification performance among all methods, with mIoU, m$F1$, and OA reaching 81.67%, 89.65%, and 92.43%, respectively. ABCNet, MANet, and A^2-FPN introduce attention mechanisms to enhance segmentation performance. With the assistance of a bilateral architecture and dot-product attention mechanism, ABCNet simultaneously captures spatial details in low-level features and global high-level contextual information. It improves 0.61% in mIoU, 0.4% in m$F1$, and 0.17% in OA compared to DeepLab V3+. MANet employs a kernel attention mechanism to extract global contextual dependencies and utilizes self-attention modules to hierarchically aggregate relevant contextual features, achieving better recognition of trees and cars than ABCNet. A^2-FPN introduces an attention aggregation module in the FPN. Its recognition capabilities for impervious surfaces, buildings, low vegetation, and trees outperform MANet's. BSNet and MCCANet enhance the utilization of boundary information. Specifically, BSNet employs dynamic mixed gradient convolution to encode boundary information, enhancing boundary extraction and improving the recognition accuracy of small objects. Experimental results show an improvement in the recognition accuracy of the car category. MCCANet uses multiscale channelwise cross attention to capture multiscale features and employs a boundary supervision module to restore boundary contours, resulting in better segmentation performance than A^2-FPN. DC-Swin, CMTFNet, BANet, and UnetFormer adopt an encoder–decoder network architecture that combines CNN and transformer, with the transformer

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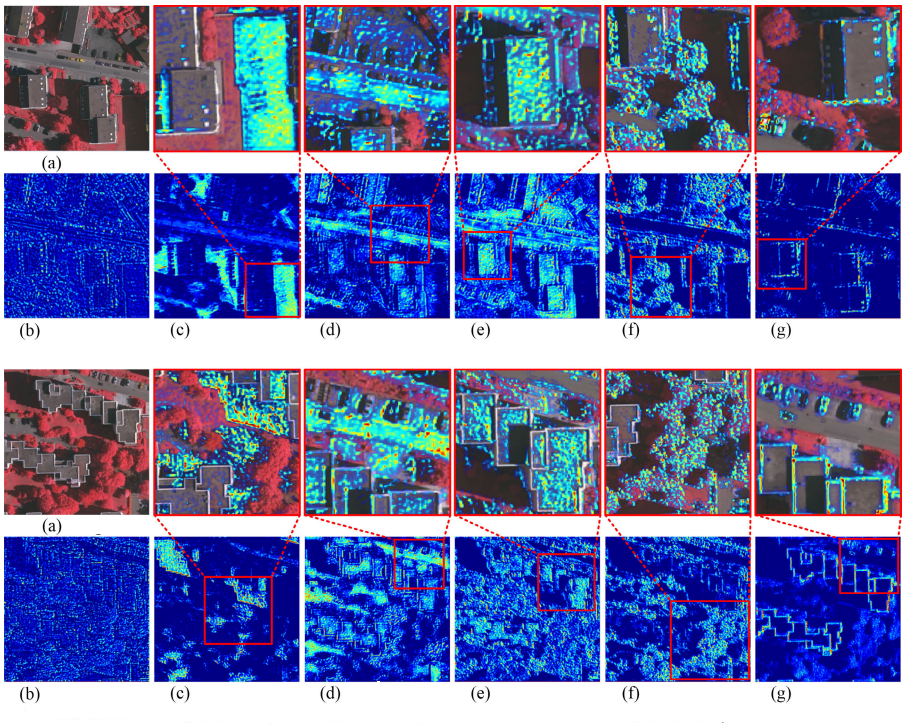


Fig. 12. Visualization results of feature maps from different GDSMs in the ResBlock Stage 1 of the encoder. (a) Input image. (b) Original feature map. (c) Feature map output from $GDSM_1$, highlighting the shadow regions. (d) Feature map output from $GDSM_2$, emphasizing the smooth road surfaces. (e) Feature map output from $GDSM_3$, accentuating the rooftops and grasslands. (f) Feature map output from $GDSM_4$, emphasizing the trees. (g) Feature map output from $GDSM_5$, highlighting the boundaries of cars and buildings.

TABLE V
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE VAIHINGEN DATASET

Method	F1 per category (%)					mIoU (%)	mF1 (%)	OA (%)
	ImpSurf	Building	LowVeg	Tree	Car			
DeepLab V3+ [74]	96.45	94.76	82.56	88.95	85.54	81.67	89.65	92.43
ABCNet [32]	96.57	94.99	83.19	88.91	86.58	82.28	90.05	92.60
MANet [43]	96.49	94.98	82.95	89.14	86.65	82.28	90.04	92.59
A^2-FPN [44]	96.75	95.50	83.73	89.38	86.55	82.84	90.38	92.98
BSNet [20]	96.47	95.07	83.40	89.19	86.89	82.52	90.20	92.68
MCCANet [45]	96.76	95.55	83.85	89.44	86.81	83.00	90.48	93.03
DC-Swin [56]	96.58	95.02	83.45	89.26	86.22	82.38	90.11	92.74
CMTFNet [52]	96.66	95.07	83.74	89.30	86.42	82.58	90.24	92.83
BANet [59]	96.60	95.13	84.05	89.52	87.92	83.28	90.64	92.93
UNetFormer [61]	96.61	95.11	83.64	89.17	86.35	82.49	90.18	92.76
GDGNet (Ours)	**96.85**	**95.57**	**84.18**	**89.68**	**88.14**	**83.64**	**90.88**	**93.16**

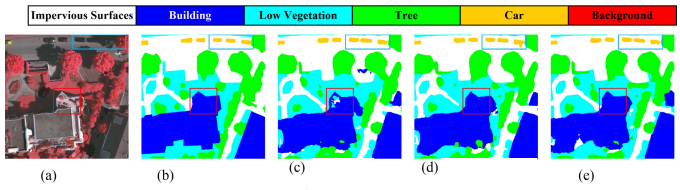


Fig. 13. Visual comparison of different model segmentation results. (a) Image. (b) Ground truth. (c) GLCF without global branch. (d) GLCF without local branch. (e) GDGNet.

designed to capture long-range dependencies in remote sensing images. DC-Swin utilizes the Swin Transformer as the encoder, achieving mIoU, m$F1$, and OA of 82.38%, 90.11%, and 92.74%, respectively. However, the initial tokenization process of the transformer may compromise the intrinsic details of objects, causing the loss of boundaries and details, which results in lower $F1$-scores for the car category compared to BSNet and MCCANet. CMTFNet and UnetFormer use CNN as the encoder to extract multiscale features and employ the transformer as the decoder, achieving better classification performance than DC-Swin. BANet adopts a dual-branch structure, where the transformer branch captures long-range relationships and the convolutional branch extracts texture features, effectively fusing dependency features and texture features using the linear attention mechanism. BANet achieves mIoU and m$F1$ of 83.28% and 90.64%, respectively, ranking second.

For the proposed GDGNet, the mIoU, m$F1$, and OA achieve the highest values of 83.64%, 90.88%, and 93.16%,

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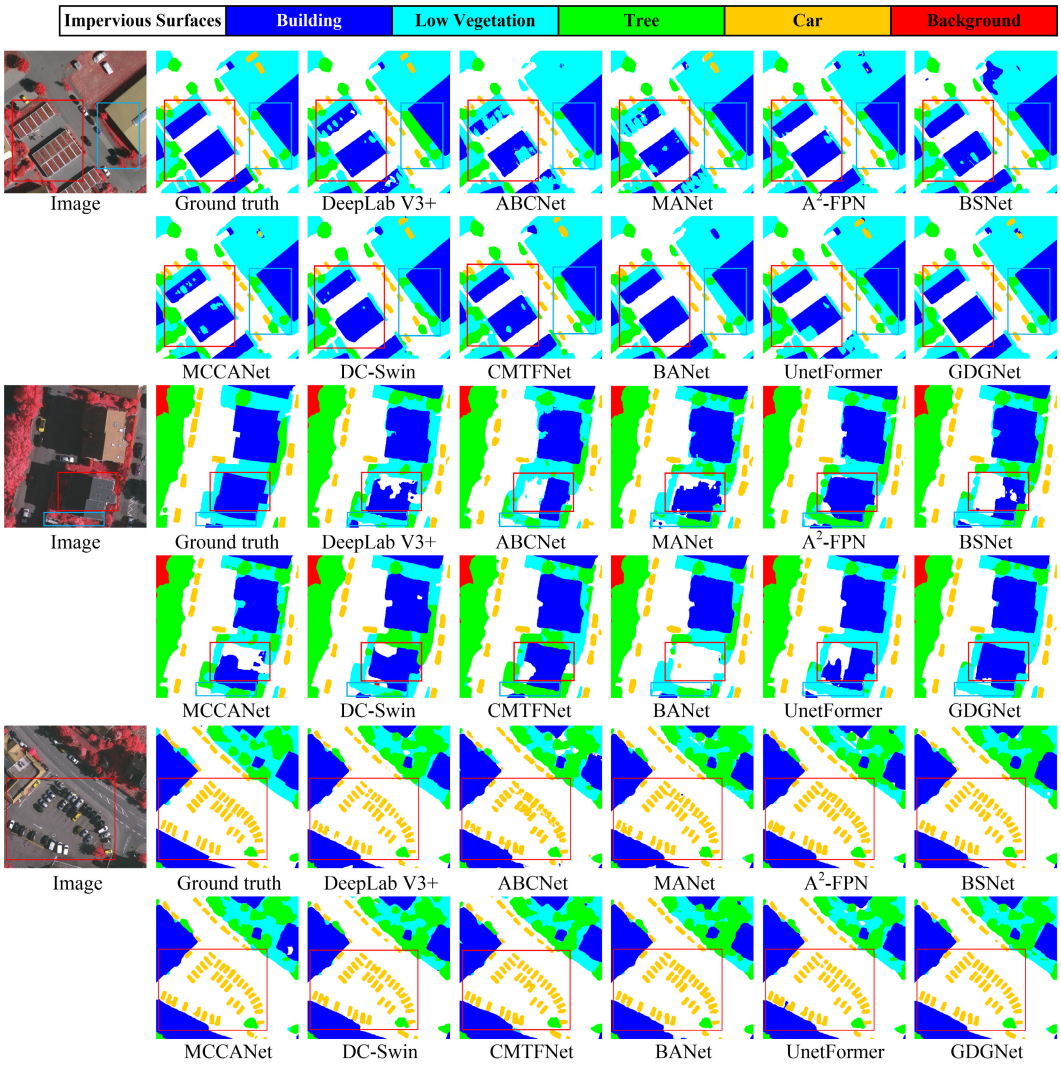


Fig. 14. Visualization comparison of segmentation maps of different methods on the Vaihingen dataset.

TABLE VI
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE POTSDAM DATASET

Method	F1 per category (%)					mIoU (%)	mF1 (%)	OA (%)
	ImpSurf	Building	LowVeg	Tree	Car			
DeepLab V3+ [74]	91.49	94.26	86.04	86.74	94.79	83.13	90.67	89.10
ABCNet [32]	92.19	94.76	86.13	87.31	93.78	83.39	90.83	89.66
MANet [43]	93.02	95.76	86.76	87.95	95.12	84.93	91.72	90.51
A^2-FPN [44]	93.28	95.90	86.74	87.96	95.32	85.14	91.84	90.67
BSNet [20]	92.75	95.70	86.60	87.76	95.79	84.95	91.72	90.33
MCCANet [45]	93.08	95.81	86.66	87.51	95.71	85.01	91.75	90.50
DC-Swin [56]	92.85	95.63	86.65	87.82	95.24	84.78	91.64	90.39
CMTFNet [52]	92.93	95.68	86.94	87.79	95.43	84.98	91.75	90.52
BANet [59]	93.12	95.96	86.92	**88.49**	94.86	85.16	91.87	90.72
UNetFormer [61]	92.93	95.59	86.84	87.82	95.52	84.96	91.74	90.46
GDGNet (Ours)	**93.29**	**96.28**	**87.03**	88.02	**95.89**	**85.60**	**92.10**	**90.78**

respectively. Clues to the segmentation performance improvements can be found in Fig. 14, which illustrates examples of predicted results on the Vaihingen dataset. One of the reasons for misclassifications could be the remarkably similar color features shared by objects of different categories. For example, as highlighted by the blue boxes in the first four rows of Fig. 14, low vegetation and trees exhibit similar reddish hues in the composite image composed of red, green, and near-infrared bands, making differentiation challenging. GDGNet employs the HGE module to enhance gradient features, amplifying the textural differences between trees and low vegetation, thereby obtaining more discriminative features. In contrast, GDGNet discriminates more accurately between trees and low vegetation. Furthermore, in the first and second rows of Fig. 14, the color and texture features on the surfaces of buildings resemble those of low vegetation. Segmentation results

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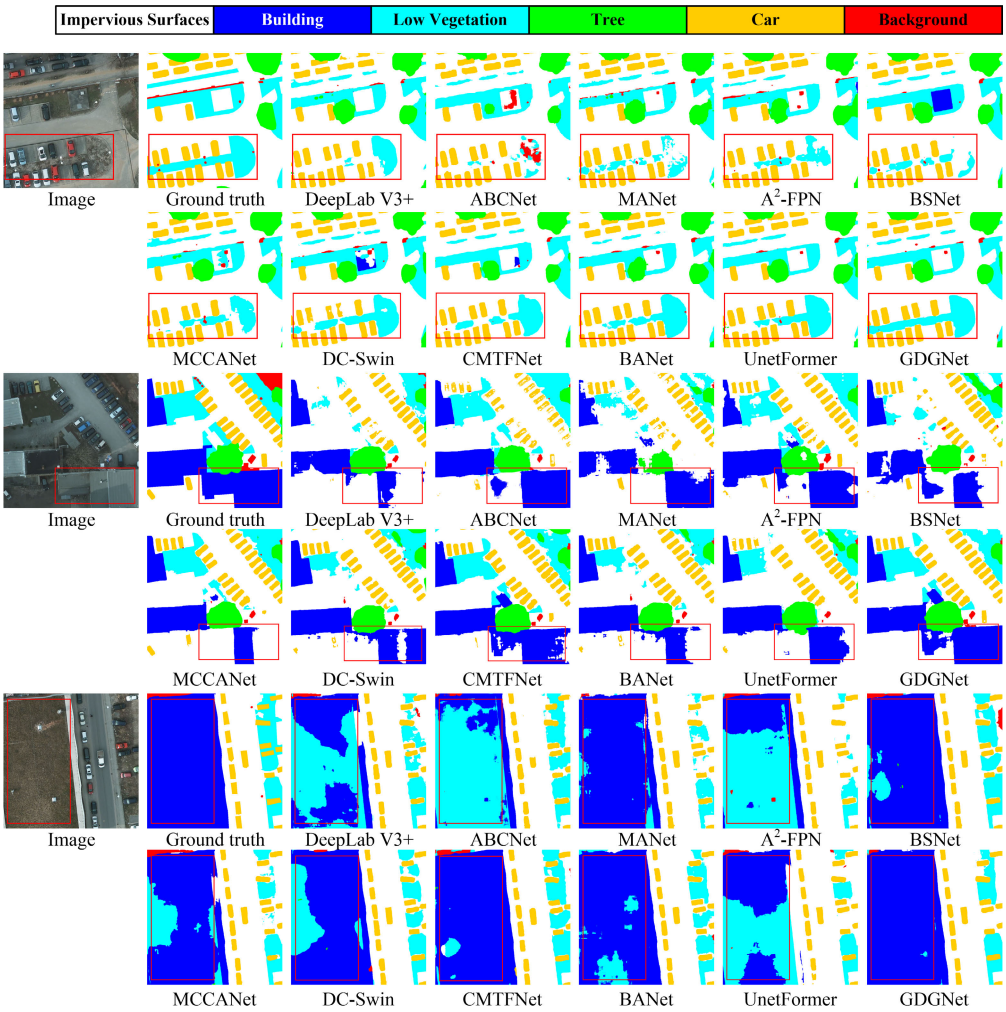


Fig. 15. Visualization comparison of segmentation maps of different methods on the Potsdam dataset.

indicate that other comparative methods erroneously classify buildings as low vegetation. With the assistance of the GLCF module, GDGNet effectively utilizes both local detailed features and global contextual information, ensuring the integrity of large-scale building objects and successfully addressing this issue. Moreover, shadows make it challenging to distinguish between buildings and impervious surfaces in the third and fourth rows of Fig. 14. GDGNet employs gradient attention masks corresponding to low gradient values to guide the model in increasing attention to shadow regions, thereby improving segmentation. For cars, their small scale and dense arrangement pose challenges for classification, particularly at boundaries. The cars predicted by the previous methods are incomplete or interconnected, while GDGNet achieves more precise car segmentation by focusing on boundary regions using gradient attention masks corresponding to high-gradient values.

2) Comparisons on the Potsdam Dataset: The segmentation performance of various comparative methods on the Potsdam dataset is presented in Table VI. ABCNet, MANet, A^2-FPN, BSNet, and MCCANet continue demonstrating the advantages of attention mechanisms and boundary information extraction in remote sensing image segmentation, achieving better performance than the conventional DeepLab V3+ method designed for natural image segmentation. Benefiting from the transformer's expression of global contextual information and the effective exploration of local texture information by the convolutional networks, CMTFNet, BANet, and UnetFormer maintain excellent segmentation performance on the Potsdam dataset. The proposed GDGNet achieves the highest mIoU, m$F1$, and OA scores of 85.60%, 92.10%, and 90.78%, respectively, with improvements of 0.44%, 0.23%, and 0.06% over BANet. Furthermore, GDGNet achieves the highest $F1$-scores for impervious surfaces, buildings, low vegetation, and cars.

Fig. 15 presents some visual segmentations on the Potsdam dataset. As shown in the first and second rows of Fig. 15, the previous methods exhibit significant misclassification between low vegetation and impermeable surfaces. One possible reason is the high intraclass variation within the low vegetation category, where sparse vegetation distribution affects its color and texture representation. However, GDGNet enhances the representation of gradient information, better exploiting the texture differences between low vegetation and impervious surfaces. Consequently, low vegetation is effectively differentiated from impervious surfaces. For large-scale objects, GDGNet maintains outstanding segmentation performance. As illustrated in the red-boxed regions of the last four rows of Fig. 15, the previous methods misclassify certain areas of buildings as either low vegetation or impervious surfaces. In contrast, GDGNet

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TABLE VII
COMPARISONS WITH STATE OF THE ARTS ON PARAMETERS, FLOPS, AND INFERENCE SPEED

Method	Parameters (Mb)	FLOPs (Gbps)	Speed (FPS)
DeepLab V3+ [74]	43.01	138.64	18.30
ABCNet [32]	14.04	63.05	**21.73**
MANet [43]	35.86	310.68	4.31
A^2-FPN [44]	26.03	361.78	4.12
BSNet [20]	21.82	117.50	15.72
MCCANet [45]	42.35	**416.62**	4.15
DC-Swin [56]	**106.86**	183.82	8.26
CMTFNet [52]	30.56	131.40	12.54
BANet [59]	12.73	52.71	18.75
UNetFormer [61]	24.24	96.71	19.16
GDGNet (Ours)	19.60	221.12	7.25

enhances the gradient information of building boundaries and employs high-gradient attention masks to strengthen the focus on building contour regions. Additionally, it combines local texture details with global contextual information to ensure the integrity of large-scale object extraction. Regarding small-scale objects such as cars, our method still exhibits outstanding recognition capabilities, capable of predicting clear boundaries and complete cars.

E. Efficiency Analysis

To evaluate the computational efficiency of GDGNet, we compared its number of parameters, floating-point operations per second (FLOPs), and inference speed (FPS) for a single patch of size 1024 × 1024 with several state-of-the-art methods in the comparative experiments. All experiments were conducted using the PyTorch framework on an NVIDIA TITAN Xp GPU. As shown in Table VII, the parameters and FLOPs of GDGNet are 19.60 M and 221.12 G, respectively. Although the FLOPs are higher than those of lightweight models such as BANet (52.71 G) and UNetFormer (96.71 G), GDGNet remains competitive compared to more complex models such as MANet and MCCANet. GDGNet achieves an inference speed of 7.25 frames/s, outperforming A^2-FPN (361.78 G), MANet (310.68 G), and MCCANet (416.62). While it operates at a lower FPS than lightweight models such as BANet (18.75 frames/s) and UNetFormer (19.16 frames/s), GDGNet effectively balances computational cost and accuracy. Future work will further optimize the architecture to reduce GFLOPs while maintaining accuracy.

V. CONCLUSION

This article introduces a novel GDGNet to handle the semantic segmentation of high-resolution remote sensing images. This method employs the HGE module to simultaneously encode and enhance boundary and texture gradient features, alleviating the insufficient extraction of texture information. In addition, a novel approach is proposed that utilizes gradient information to decouple low-level features, optimizing the representation of these features. Considering the difference between boundary and texture gradient features, we designed GDM and HGA to further improve the feature representation in specific regions and solve the potential feature entanglement problem in the low-level features. By the GLCF module, GDGNet could capture both local and global contextual information while adaptively fusing low- and high-level features, resulting in more discriminative fused features. The ablation experiments validate the effectiveness of each component of the proposed method. Additionally, extensive comparison experiments demonstrate that GDGNet achieves state-of-the-art performance.

Since GDGNet can generate GDMs to guide the model in refining low-level features without sample labels, future research will focus on applying GDGNet to unsupervised and semisupervised remote sensing image segmentation tasks.

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